

# BIKES: A Mobile Networked Music Instrument in Interdisciplinary Research and Education

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## ABSTRACT

*This paper describes how a mobile, networked instrument for music and sound art is used as a platform for interdisciplinary research and creative practice in higher education. The long-term project BIKES provides students with the opportunity to engage with real-world challenges by combining music technology, experimental composition, and industrial design. Project activities include interactive installations and sound rides, iterative development of hardware and software, as well as the design and fabrication of a new prototype for exhibition contexts. After its first year, BIKES demonstrates how the multifaceted nature of a modular instrument can facilitate collaborative work and increase the visibility of student-led research and development.*

## 1. INTRODUCTION

### 1.1 The Interdisciplinary Nature of BIKES

Projects in the field of computer music and sound art are inherently interdisciplinary. Instruments must be designed and built, often involving digital fabrication, circuit design, and hardware prototyping. Software must be developed using a range of programming languages and paradigms. Finally, the processes of composition and performance require not only creative thinking, but also critical contextualization and methods drawn from the humanities. In an educational context, this interdisciplinarity presents both significant challenges and substantial opportunities.

With BIKES, we employ a networked musical instrument as a platform for interdisciplinary research, artistic practice, and pedagogical experimentation, exploring mobile, spatially situated performances and installations, as shown in Figure 1. In the context of this project, up to eight undergraduate students are engaged, co-led by graduate students and faculty. With most participating students majoring in music technology, the BIKES project positions this field as a hub discipline that coordinates methods from interaction

design, industrial design, fabrication, embedded systems, and sound studies. In doing so, the project foregrounds learning through practice, encouraging students to engage with unfamiliar disciplinary perspectives and to reflect critically on their own design and artistic processes.



**Figure 1:** Visitors interacting with BIKES at an installation  
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Through project-based exploration, interdisciplinary design work supports a collaborative learning environment in which students can integrate technical and artistic approaches in the development of systems such as BIKES. By integrating disciplines with differing pedagogical frameworks, the project combines linear knowledge structures common in STEM fields with dialogue-driven learning approaches often found in arts and non-STEM education [1]. This interdisciplinary structure aligns with findings that students benefit from learning environments that blend independent technical work with interactive, hands-on collaboration [1].

This paper highlights the interdisciplinarity of BIKES by outlining its development across multiple project phases and illustrating the diversity of artistic and technical practices involved.

## 1.2 Background and Context

Building on the networked music system *SPRAWL* [2], BIKES is conceived as a mobile, networked platform for artistic expression and interaction, aiming to increase accessibility by bringing sound art to new locations and enabling new forms of composition and performance. Mobile sound practices in public spaces have been explored in a range of artistic projects that treat physical movement as a constitutive element of the work. Earlier work, such as Max Neuhaus’ *Listen*, established walking and mobility as integral to listening, positioning movement through public space as a central artistic concern [3]. More recently, Kaffe Matthews’ *Sonic Bikes* employs bicycle-based mobility in urban environments as a compositional and experiential strategy, foregrounding movement through the city as part of the musical structure [4].

Although BIKES is intended as a flexible platform for various forms of sound art and music, one of its primary use cases is the creation of urban soundscapes, realized either as stationary installations or through movement in public space. In developing a soundscape-oriented interactive system, the sound design of BIKES draws on R. Murray Schafer’s descriptive framework, which distinguishes between *Keynote* sounds, *Signals*, and *Soundmarks* as functional components of an acoustic environment [5]. Within this framework, Keynote sounds form a continuous sonic ground, often shaped by environmental or geographic conditions, while Signals demand focused attention as foreground elements. Soundmarks denote distinctive sounds associated with the acoustic identity of a particular place or community [5].

Beyond Schafer’s taxonomy, later soundscape practices have emphasized situated listening, participation, and critical engagement with urban environments, particularly in response to social, political, and ecological contexts [6, 7]. Taken together, these perspectives informed the design of a subsequent iteration of BIKES, which employs layered and interactive sonic elements to construct responsive soundscape environments that evolve in relation to place and movement.

The progression of BIKES can be categorized into three distinct phases. Phase 1, introduced in Section 2, focused on the design and development of a first working prototype and delivered a proof of concept for the complete instrument through its initial activations in spring 2025. Section 3 reports on Phase 1.1, which introduced several improvements to the hardware and software of each bicycle, followed by professional activations and increased outreach in summer and fall 2025. The ongoing Phase 2 of BIKES is dedicated to the implementation of a new prototype based on a custom-designed loudspeaker enclosure. Section 4 introduces first results from this phase, alongside documentation of an exhibition of the project.



Figure 2: BIKES in the laboratory during development.

## 2. PHASE 1

### 2.1 Hardware and Software

BIKES is based on an electric cargo bike manufactured by Edison Bicycles, an Atlanta-based electric bike company [8]. The model features a 750 W motor that supports both pedal assist and throttle control. A rear cargo section provides sufficient structural support to house the audio hardware while maintaining mobility. The payload capacity of 181 kg allows the integration of professional audio equipment without compromising range or road speed. Four of these bikes are equipped with music technology to form the complete instrument used in Phase 1. Figure 2 shows the bikes in the laboratory during the development stage.

Each bike runs on a Raspberry Pi 5 computer [9], mounted in a custom 3D-printed enclosure fitted to the handlebar. This enclosure also holds a 10-inch touchscreen that serves as the main user interface. One battery-powered loudspeaker per bike, along with an external audio interface and additional peripherals, is mounted on the rear cargo section. Further details of the hardware setup are described in a previous publication [8].

The networked music system *SPRAWL* informed parts of the BIKES design [2]. While *SPRAWL* relied on wired Ethernet connections to implement a fully connected audio routing system for ensemble interaction, BIKES uses a WiFi-based network for the exchange of control data rather than audio signals. In its first iteration, BIKES operated with a centralized network topology, in which each bicycle node connected to a single central router carried by one bicycle. While this centralized topology was sufficient for an initial proof of concept, the intended direction was to transition to a decentralized mesh architecture in order to reduce single points of failure and network dropouts.

The networking capabilities constitute a core aspect of the system, both in terms of compositional possibilities and technological demands. Communication between the nodes is implemented using OSC (Open Sound Control) [10]. Additionally, *p2psc*<sup>1</sup> is used to simplify communication between the bicycles. This Python-based tool allows each node to register with a unique identifier and to

<sup>1</sup> <https://github.com/l42i/p2psc>

subscribe to specific OSC paths, enabling flexible message exchange between devices [8].

SuperCollider was used for audio signal processing and the design of graphical user interfaces (GUIs) in the first phase. Each Raspberry Pi runs an independent SuperCollider server and language instance, with graphical user interfaces implemented in SuperCollider to provide touch-based control over audio and network processes [8].

Finally, *Ansible* is used to control and configure the instrument. This professional IT tool enables remote configuration and deployment of individual compositions across the system. Configuration is handled via SSH commands organized in playbooks, which specify ordered sequences of script operations executed on a set of hosts. An Ansible script is used to remotely start the SuperCollider processes and other software on each Raspberry Pi, eliminating the need to manually launch programs on individual nodes. This approach allows BIKES to be deployed on site without per-node configuration.

## 2.2 Activations and Proof of Concept

To demonstrate the system as a proof of concept and to evaluate its strengths and limitations, BIKES was deployed in a range of public scenarios in spring 2025. These deployments included tests of the user interface and battery life, as well as interactive installations and fully networked rides that presented the system in motion [8].

A first stress test took place at an Undergraduate Research Symposium at Georgia Tech, where a single bike was displayed and visitors interacted with the touch interface while triggering harmonic sounds over a looped background. This was followed by an interactive quadraphonic installation during Launchpad, the annual showcase at Georgia Tech’s College of Design, where the installation was exposed to thousands of visitors and demonstrated the system’s reliability, portability, and synchronization in a high-throughput public setting. Finally, the full system was evaluated in motion during the Inman Park Parade, where four fully equipped bikes performed along a two-mile route, demonstrating both a synchronized clock mode and an independent node paradigm. This activation provided the first opportunity to test and explore the potential of real-time sound control and spatialized audio in a dynamic, mobile performance context.

## 3. PHASE 1.1

### 3.1 Hardware and Software

Phase 1.1 represents the current working configuration of BIKES. In terms of the hardware design, only minor changes were made to the first working version in order to increase the robustness of the system for extended rides. These measures included additional padding for loudspeakers and peripherals, bungee cords to secure the lid of the wooden enclosure, and minor modifications to the display enclosure.

The most significant change in this phase concerns the network. Using one UniFi AC Mesh router on each bike, BIKES was upgraded to a mesh network, representing

a substantial improvement over the first iteration. This model was chosen for its extended outdoor range of 600 ft, reducing connection dropouts and latency in comparison to the previous setup. One router is assigned as the parent node connected to a network gateway, while all routers are interconnected to form a mesh. Each Raspberry Pi is connected to its associated mesh router via Ethernet, resulting in the structure shown in Figure 3.

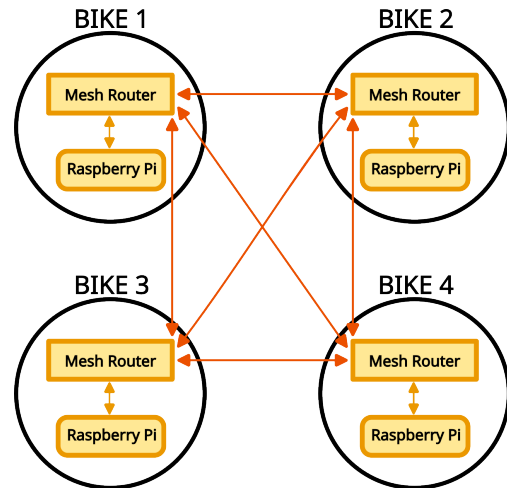


Figure 3: Updated network concept for Phase 1.1.

SuperCollider is retained as the main audio backend. Its ability to run fully scripted and headless makes it well suited for this application. The OSC communication approach is also retained, relying on direct communication between the SuperCollider instances and p2ps. To introduce new possibilities for interactive design, Python’s Pygame library is used to create graphical user interfaces for installations in Phase 1.1. This portable, SDL-based framework remains lightweight and accessible in an educational context, while providing access to graphics, audio, and other multimedia hardware across operating systems [11].

### 3.2 Activations and First Soundscape Compositions

After the proof of concept in the first phase and with improved hardware and software, BIKES was deployed in several activations in fall 2025. The project was presented as a stationary interactive installation at SITE, a multimedia arts festival at The Goat Farm in Atlanta<sup>2</sup>. This installation marked the first presentation of a distributed soundscape installation specifically designed for BIKES. The four bicycles were arranged in a rectangle in an alleyway to take advantage of the acoustics, with each bike’s speakers facing inward, as shown in Figure 4.

Following Schafer’s framework, several distinct soundscapes were designed. These ranged from a serene ocean setting to industrial scenarios and rainforest environments. Keynote sounds were played simultaneously across the four bicycles and continuously for the duration of the performance, acting as a central foundation for the soundscape. Signals and Soundmarks could be triggered by the audience through the graphical user interfaces. Each bike’s

<sup>2</sup> <https://www.thegoatfarm.info/site2025>



**Figure 4:** Setup of BIKES at SITE for an interactive installation.

display showed a simple GUI that allowed visitors to trigger sound events played back on all loudspeakers, following quasi-random patterns. The screen displayed an image representing the environment, divided into a grid of buttons. Depending on the button pressed, a single instance was played by the same bicycle, repeated instances were passed between bicycles in a predetermined pattern, or repeated instances were passed around in a random pattern. This resulted in a dense and immersive sonic environment with direct means for audience interaction.

Building on the successful presentation of an interactive installation, Phase 1.1 of BIKES was deployed as a mobile instrument during a public group bike ride held in collaboration with The Fulton County Public Art Program<sup>3</sup>, MOBB<sup>4</sup>, and Red Bike and Green<sup>5</sup>. The aim of this site-specific experience was to highlight strengths and weaknesses of Atlanta's urban environment, both in terms of acoustics and safety for bicycles and pedestrians. This capacity for critical engagement with the surrounding environment constitutes a key aspect of the instrument. More than 30 riders accompanied the instrument on a two-hour route through the inner city, temporarily occupying full traffic lanes in the manner of a Critical Mass ride [12].

Two concepts were explored in this activation. The first involved a playlist of songs by established and popular local artists, curated for the ride by the students. When in motion, all four bikes played the same stereo file per piece, synchronized via SuperCollider. Through the connection to Atlanta as a hub for music production and performance, this approach created a direct connection to the incidental audience. The compositions were easily recognized and elicited immediate positive responses from bystanders and participants along the route.

The second concept was intended for impromptu installations during the ride. The group stopped at previously selected locations to demonstrate the soundscapes introduced earlier in this subsection. Some of these locations were chosen as illustrative examples of favorable or challenging urban acoustic environments, including freeway overpasses, tunnels, intersections, and paths through parks. Other locations were added as stops due to their status as iconic landmarks in the city. The soundscapes were designed to either contrast with or complement the surrounding acoustics, encouraging critical listening among the audience.

<sup>3</sup> <https://www.fultonarts.org/publicart>

<sup>4</sup> <https://www.mobbcycling.com/>

<sup>5</sup> <https://www.redbikegreenatl.com/>

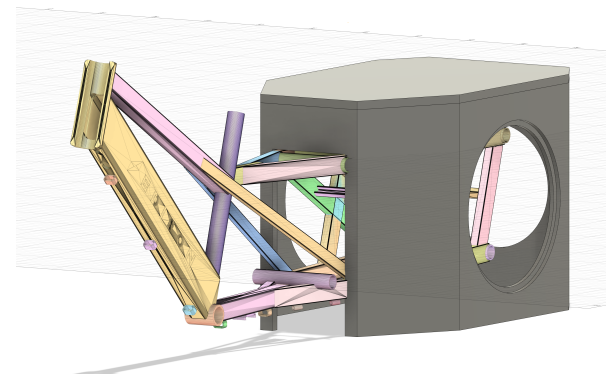
## 4. PHASE 2

### 4.1 Prototype Design and Fabrication

Following the proof of concept and initial public activations of the improved system, Phase 2 of BIKES shifted its focus toward the design and fabrication of a custom hardware solution for housing the loudspeakers and onboard technology. This transition corresponded with a shift in project activities, moving from software development and artistic exploration toward industrial design and digital fabrication.

To address hardware limitations present in the working instrument, a new, rugged loudspeaker enclosure was designed and built. With the aim of improving stability and long-term durability, a wooden frame was constructed around the rear wheel to lower the center of mass and improve rideability. Based on these design principles, a rapid prototyping framework was selected to develop an enclosure that integrates not only the system's loudspeakers but also the electronic hardware required for music generation and network interaction in future design iterations.

Crucially, prior feedback from field tests conducted in Phase 1 informed these design decisions. For example, operators of the initial prototype reported safety concerns related to a high center of gravity, which resulted in unpredictable steering at low speeds. This feedback led to the decision to position the enclosure around the rear wheel, ensuring a center of gravity as low as possible to maximize rider maneuverability while avoiding ground contact on uneven terrain. Figure 5 shows a 3D model of the basic design, which features two 15 inch woofers mounted on either side and one 12 inch woofer at the rear.



**Figure 5:** CAD Model of Phase II speaker cabinet.

Given the timing of the second design phase and the plan to present the system as a stationary object in a display context, a *looks-like* model was fabricated to communicate the artistic and material qualities of BIKES without the ability to produce sound. In this context, a *looks-like* model refers to an appearance-focused mock-up intended to convey form, scale, and material presence, whereas a *works-like* prototype prioritizes demonstrating functional behavior [13]. These distinctions from industrial design and HCI research informed the development of Phase II, with the initial iteration of BIKES serving as the *works-like* prototype [2], while the second phase presents an explanatory, non-functional artifact focused on visual and spatial communication.

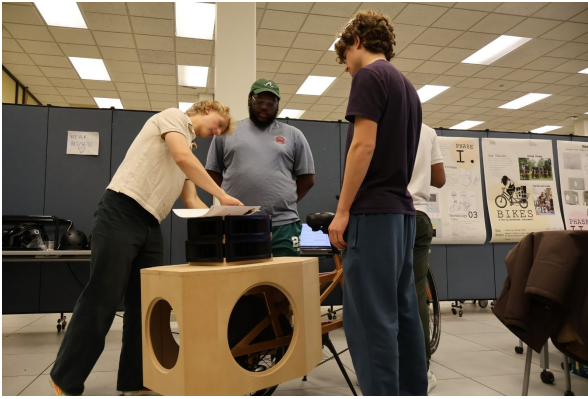


Figure 6: Phase 2 prototype manufacturing in the lab.

The design process followed a series of iterative steps consistent with research-through-design methodologies, in which design activity itself functioned as a mode of inquiry, supporting both creative exploration and team communication throughout development. First, the strengths and limitations of the original model were systematically analyzed to identify design constraints and opportunities. Based on this analysis, initial concepts were developed through hand sketches and reflective discussion, establishing a provisional understanding of the instrument's overall form. Subsequently, scale-accurate drawings were produced to examine proportions and identify inconsistencies, enabling iterative refinement prior to fabrication. These refined drawings were then translated into Computer-Aided Design (CAD) models using software such as Autodesk Fusion, which served as the basis for fabrication on an industrial Computer Numerical Control (CNC) machine, allowing the enclosure to be assembled by hand. Figure 6 shows the addition of the upper structure supporting eight 10.5 inch horn tweeters to the basic enclosure in the laboratory. The tweeters are arranged in a square configuration to facilitate omnidirectional sound projection.

The complete enclosure is coated with a weather-resistant loudspeaker finish, matching the black visual language of the cargo bicycle. Figure 7 shows the assembled prototype, highlighting the lowered center of gravity. Although this iteration constitutes a non-functional prototype, the design is intended to accommodate the necessary hardware to be developed into a fully operational system.

#### 4.2 Exhibition Concept and Design

The Phase 2 prototype introduced in Section 4.1 was specifically developed for display in a public exhibition context at Atlanta Hartsfield–Jackson International Airport, curated by Georgia Tech Arts<sup>6</sup>. Figure 8 shows an early concept sketch for the display, illustrating the project's trajectory by presenting the appearance model alongside contextual materials, including posters and a documentary video showing the operational instrument in use.

In addition to the physical prototype, the display incorporated a set of supplementary materials intended to contextualize the system for a non-specialist public. A se-

<sup>6</sup> <https://design.gatech.edu/feature/design-airport-exhibit-2025>



Figure 7: Phase 2 speaker cabinet mounted on the cargo bike.

ries of three posters presented key aspects of Phase 1 and Phase 2, emphasizing visual material and concise text blocks to communicate the project's development and design logic. To further convey the concept and scope of BIKES, a professionally produced video was developed in collaboration with Kevin Beasley. The video documents the project's motivation and use through real-time footage of the instrument in operation, complemented by short interviews with students and team members involved in its development. Animated visualizations embedded in the video illustrate the operation of the centralized mesh network within BIKES, interwoven with documentation from multiple public deployments throughout 2025. These materials were presented as part of a year-long display alongside the Phase 2 prototype at Hartsfield–Jackson Atlanta International Airport, as shown in Figure 9.

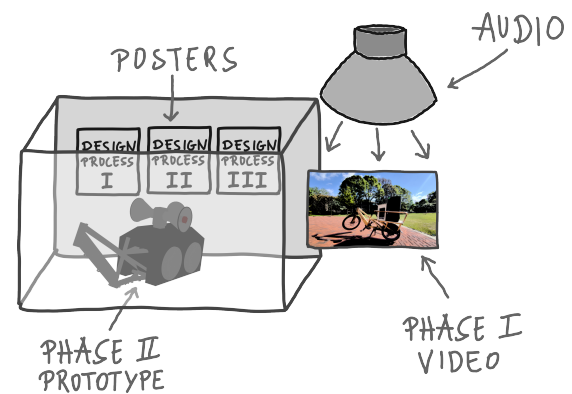


Figure 8: Original concept for the airport exhibition.

## 5. CONCLUSION

Across its first two phases, BIKES developed from a networked musical experiment into a mobile spatial audio system integrating aspects of music technology, interaction design, and industrial fabrication. Phase 1 established the technical and musical foundations of the system, including a mesh-based network architecture, distributed audio rendering, and mechanisms for audience interaction, while Phase 2 addressed questions of physical integration, durability, and visual legibility in public contexts. The



Figure 9: Exhibition with prototype and posters.

project was realized through an interdisciplinary development process in which compositional practice, software engineering, industrial design, and digital fabrication informed successive design iterations. Students from different disciplinary backgrounds contributed to the project at multiple stages, engaging in hands-on collaboration across technical, artistic, and organizational domains. As a result, BIKES functions both as a mobile ensemble for spatial sound works and as a platform for iterative design and experimentation in public space.

Future work will integrate the functional capabilities demonstrated in Phase 1 with the design principles developed in Phase 2 to realize a fully operational version of the system. In its second year, BIKES will be developed within a structured Vertically Integrated Projects (VIP) program, providing an organizational framework for continued student-led, interdisciplinary research and development.

### Acknowledgments

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